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In [2]: import os
os.environ["TF_CPP_MIN_LOG_LEVEL"] = "2"

import pandas as pd, numpy as np
import matplotlib.pyplot as plt
from tensorflow import keras
from tensorflow.keras import layers
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

FILE = r"D:\Datascience-Machinelearning\Datasets\titanic.csv"

df = pd.read_csv(FILE)[["Survived", "Pclass", "Sex", "Age", "Fare", "SibSp", "Parch"]].copy()
df["Age"] = df["Age"].fillna(df["Age"].mean())
df["Sex"] = df["Sex"].map({"male": 0, "female": 1})
df = df.dropna()

X, y = df.drop(columns="Survived"), df["Survived"]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

sc = StandardScaler()
X_train = sc.fit_transform(X_train).reshape(-1, 6, 1)  # 6 features as a 1D sequence
X_test = sc.transform(X_test).reshape(-1, 6, 1)

model = keras.Sequential([
    layers.Input(shape=(6, 1)),
    layers.Conv1D(32, 2, activation="relu"),
    layers.Conv1D(64, 2, activation="relu"),
    layers.Flatten(),
    layers.Dense(32, activation="relu"),
    layers.Dense(1, activation="sigmoid"),
])
model.compile(optimizer="adam", loss="binary_crossentropy", metrics=["accuracy"])
model.fit(X_train, y_train, epochs=30, batch_size=32, validation_split=0.1, verbose=0)

loss, acc = model.evaluate(X_test, y_test, verbose=0)
print("Test accuracy:", acc)
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pred = (model.predict(X_test) > 0.5).astype(int).ravel()
cm = confusion_matrix(y_test, pred)
print("Confusion Matrix:\n", cm)

ConfusionMatrixDisplay(cm, display_labels=["Died", "Survived"]).plot(cmap="Blues", values_format="d")
plt.title(f"CNN Confusion Matrix - Titanic (Accuracy: {acc:.3f})")
plt.savefig("confusion_matrix_cnn_titanic.png", dpi=150, bbox_inches="tight")
plt.show()

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Test accuracy: 0.8100558519363403

6/6 ————— 0s 18ms/step

Confusion Matrix:

[[93 12]

[22 52]]



